

Linear regression Lab

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Part 1

(Q1) Try changing the value of standard deviation (sd) in the next command

How do the data points change for different values of standard deviation?

Increasing the standard deviation increases the amount of noise in the data.

When sd is small, the data points lie close to a straight line.

As sd increases, the points become more scattered and the linear relationship becomes less clear.

(Q2) How are the coefficients of the linear model affected by changing the value of standard deviation in Q1?

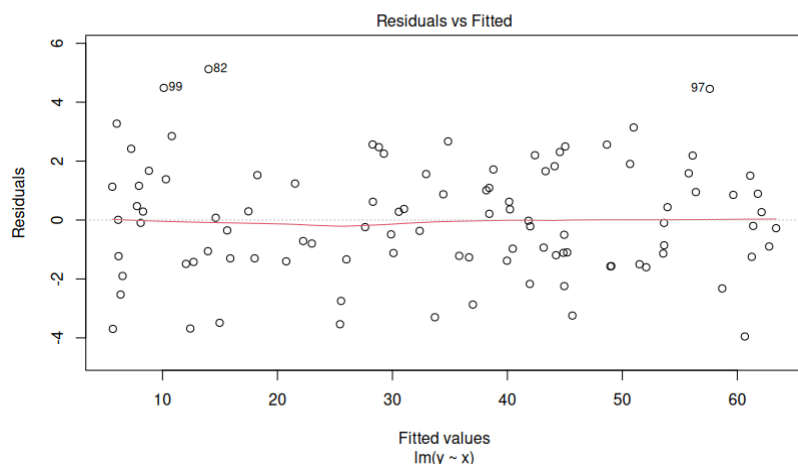
When the standard deviation increases, the estimated coefficients become less accurate and deviate more from the true values. For example, when $sd = 2$, the intercept and slope are 5.128 and 6.031, which are very close to the true values (5 and 6). However, when $sd = 5$, they change to 5.349 and 5.850, showing more deviation. This means that higher noise leads to less precise coefficient estimates, although they still remain around the true values.

(Q3) How is the value of R-squared affected by changing the value of standard deviation in Q1?

When the standard deviation increases, the R-squared value decreases. For example, when $sd = 2$, the R-squared is 0.988, indicating an excellent fit. However, when $sd = 5$, the R-squared drops to 0.919, showing a weaker fit. This happens because increasing the standard deviation adds more noise to the data, making the linear relationship less clear and reducing how well the model explains the variation in the data.

(Q4) What do you conclude about the residual plot? Is it a good residual plot?

The residuals are randomly distributed around zero, indicating a good model fit with no clear pattern in the errors.

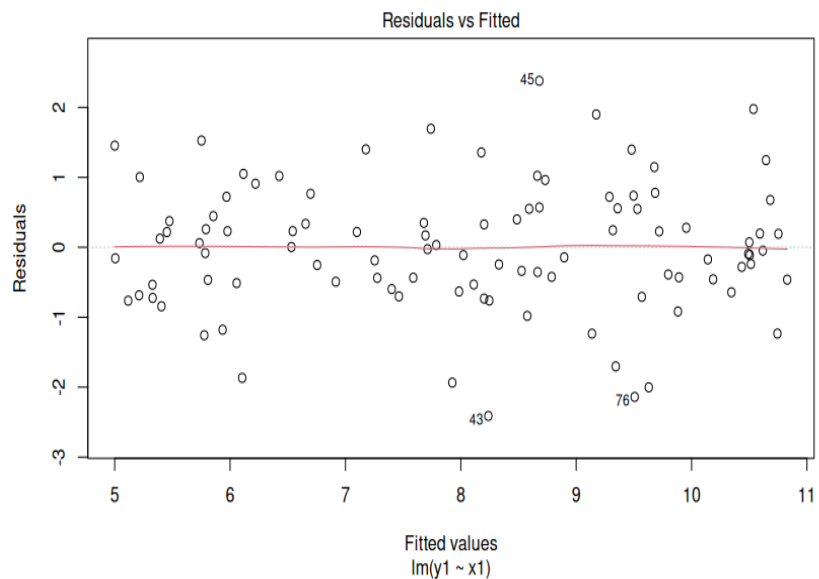


Part 2

(Q5) What do you conclude about the residual plot? Is it a good residual plot?

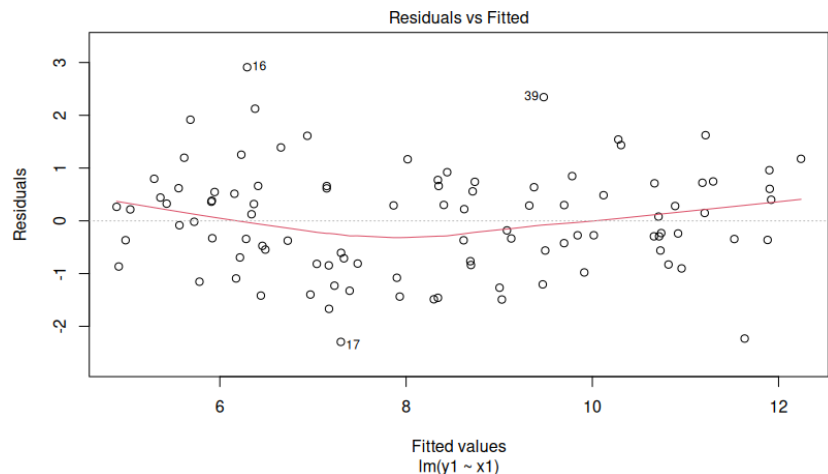
Yes, this is a good residual plot.

The points are randomly scattered around the zero line with roughly equal vertical spread across all fitted values. This indicates the linear regression model fits well and satisfies the key assumptions of linearity and constant variance.



(Q6) Now, change the coefficient of the non-linear term in the original model for (A) training and (B) testing to a large value instead. What do you notice about the residual plot?

When the coefficient of the nonlinear term is increased from 0.1 to 2, the residual plot becomes much worse. The residuals show a clear curved pattern, with some points falling far below or above zero. This indicates that the linear model cannot capture the stronger nonlinear relationship, and the errors are now larger and systematically patterned, confirming that the linear model is inappropriate for this data.



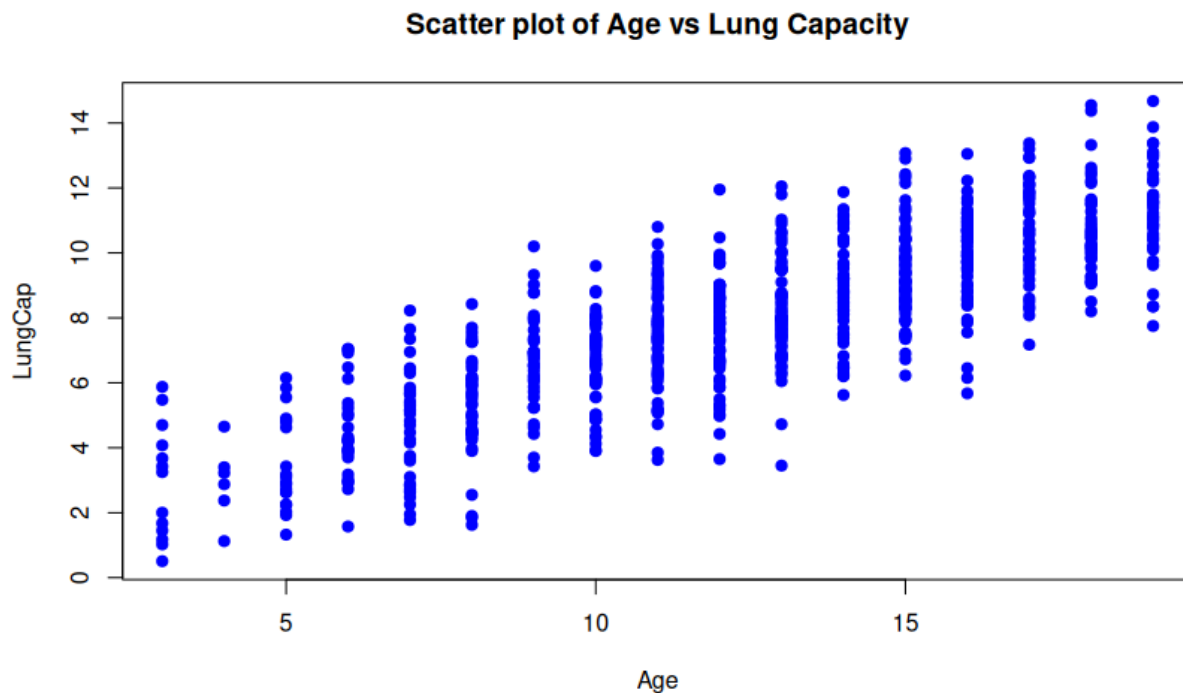
Part 3

(Q7) Import the dataset LungCapData.tsv. What are the variables in this dataset?

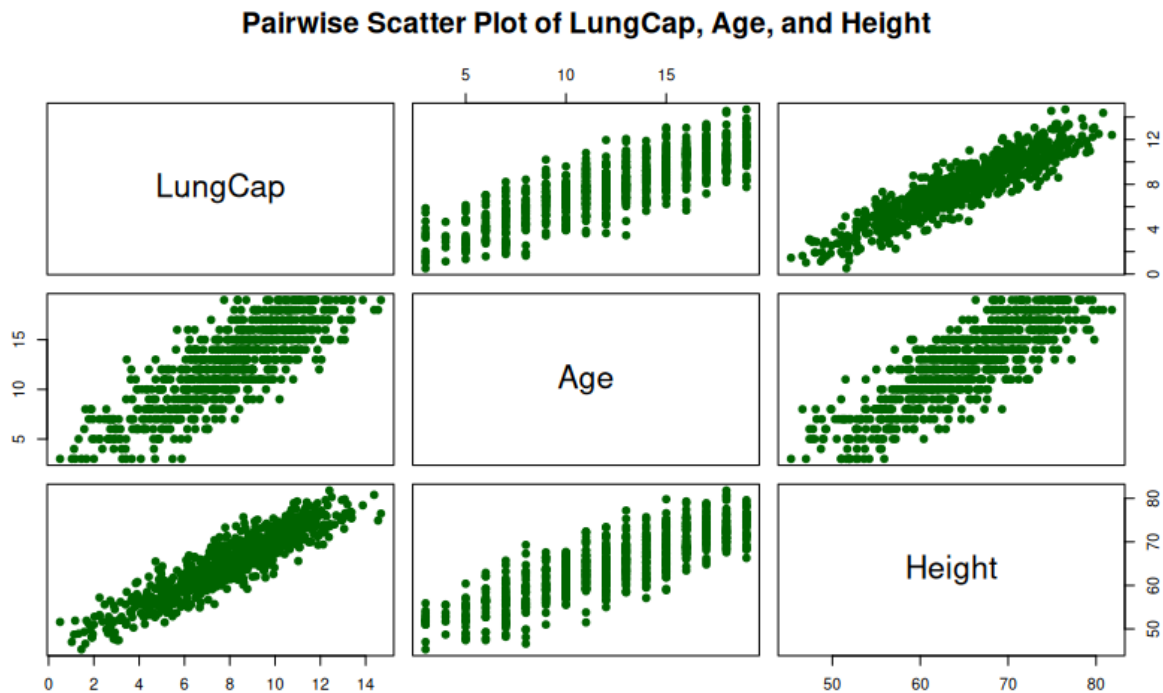
The dataset LungCapData.tsv contains 725 observations and 6 variables:

1. LungCap
2. Age
3. Height
4. Smoke
5. Gender
6. Caesarean

(Q8) Draw a scatter plot of Age (x-axis) vs. LungCap (y-axis). Label x-axis "Age" and y-axis "LungCap"



(Q9) Draw a pair-wise scatter plot between Lung Capacity, Age and Height.



(Q10) Calculate correlation between Age and LungCap, and between Height and LungCap.

Correlation between Age and LungCap: 0.8196749

Correlation between Height and LungCap: 0.9121873

(Q11) Which of the two input variables (Age, Height) are more correlated to the dependent variable (LungCap)?

Both Age and Height are positively correlated with LungCap, but Height has a stronger correlation 0.912 compared to Age 0.820. This indicates that Height is more closely related to Lung Capacity than Age in this dataset.

(Q12) Do you think the two variables (Height and LungCap) are correlated ? Why ?

Yes, Height and LungCap are strongly positively correlated = 0.912. This makes sense because taller individuals generally have larger lungs, so as height increases, LungCap also tends to increase.

(Q13) Fit a liner regression model where the dependent variable is LungCap and use all other variables as the independent variables

```
# Fit a multiple linear regression model
model_all <- lm(LungCap ~ Age + Height + Smoke + Gender + Caesarean, data = lung)
```

#(Q14) Show a summary of this model

Call:

```
lm(formula = LungCap ~ Age + Height + Smoke + Gender + Caesarean,
    data = lung)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.3388	-0.7200	0.0444	0.7093	3.0172

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.32249	0.47097	-24.041	< 2e-16 ***
Age	0.16053	0.01801	8.915	< 2e-16 ***
Height	0.26411	0.01006	26.248	< 2e-16 ***
Smokeyes	-0.60956	0.12598	-4.839	1.60e-06 ***
Gendermale	0.38701	0.07966	4.858	1.45e-06 ***
Caesareanyes	-0.21422	0.09074	-2.361	0.0185 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.02 on 719 degrees of freedom

Multiple R-squared: 0.8542, Adjusted R-squared: 0.8532

F-statistic: 842.8 on 5 and 719 DF, p-value: < 2.2e-16

(Q15) What is the R-squared value here ? What does R-squared indicate?

The R-squared value of the model is 0.854, which means that approximately 85.4% of the variation in LungCap is explained by the independent variables (Age, Height, Smoke, Gender, and Caesarean). A higher R-squared indicates that the model fits the data well and the predictors explain most of the variability in LungCap.

(Q16) Show the coefficients of the linear model. Do they make sense?
If not, which variables don't make sense? What should you do?

The coefficients of the model show that Age and Height have positive effects on LungCap, while Smoking and Caesarean have negative effects, and being male increases LungCap. These results make sense logically, as older and taller individuals and males generally have larger lungs, while smoking tends to reduce lung capacity. Overall, the coefficients align with expectations, indicating the model is reasonable.

```
> #If not, which variables don't make sense? What should you  
> summary(model_all)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.3224856	0.47097384	-24.040583	3.518644e-94
Age	0.1605296	0.01800726	8.914715	3.988466e-18
Height	0.2641128	0.01006215	26.248142	5.049675e-107
Smokeyes	-0.6095592	0.12597708	-4.838652	1.600357e-06
Gendermale	0.3870117	0.07965729	4.858459	1.452909e-06
Caesareanyes	-0.2142182	0.09073684	-2.360873	1.849787e-02

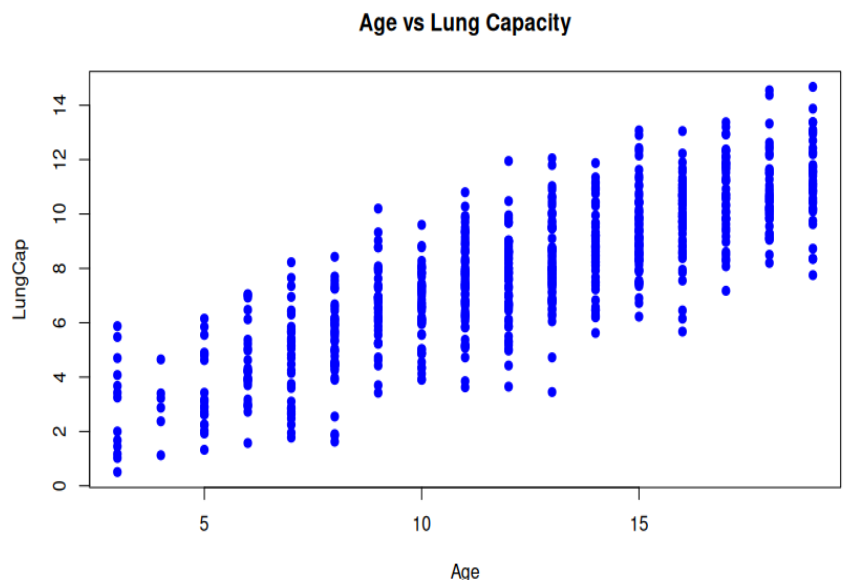
```
>
```

(Q17) Redraw a scatter plot between Age and LungCap. Display/Overlay the linear model (a line) over it.

Note (1) : A warning will be displayed that this function will display only the first two coefficients in the model. It's OK.

Note (2) : If you are working correctly, the line will not be displayed on the plot. Why?

When plotting Age vs LungCap and using `abline()`, no line appears because `abline()` can only draw a line for a simple linear regression with a single predictor. Since `model_all` is a multiple regression with several predictors (Age, Height, Smoke, Gender, Caesarean), `abline()` cannot correctly represent the fitted values in two dimensions. The warning is expected, and it is normal that the line is not displayed.



(Q18) Repeat Q13 but with these variables Age, Smoke and Cesarean as the only independent variables.

```
> model_reduced <- lm(LungCap ~ Age + Smoke + Cesarean, data = lung)
> summary(model_reduced)

Call:
lm(formula = LungCap ~ Age + Smoke + Cesarean, data = lung)

Residuals:
    Min       1Q   Median       3Q      Max
-4.7428 -1.0580 -0.0490  0.9985  4.2319

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.10867    0.18419   6.019 2.79e-09 ***
Age          0.55617    0.01439  38.639 < 2e-16 ***
Smokeyes    -0.64310    0.18681  -3.443 0.000609 ***
Caesareanyes -0.14603    0.13468  -1.084 0.278610
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.514 on 721 degrees of freedom
Multiple R-squared:  0.6778,    Adjusted R-squared:  0.6764
F-statistic: 505.5 on 3 and 721 DF,  p-value: < 2.2e-16

> |
```

##(Q19) Repeat Q16, Q17 for the new model. What happened?

After fitting the reduced model with Age, Smoke, and Cesarean as predictors, Age remains a strong positive predictor of LungCap, and Smoking has a negative effect, while Cesarean has a small, non-significant effect. Compared to the full model, the coefficients changed slightly and the overall fit is slightly worse, as expected. The red regression line now appears on the scatter plot because the model has only one numeric predictor (Age), which allows `abline()` to draw the line. Residuals are slightly larger, indicating that removing variables like Height and Gender reduces the model's explanatory power, but the main relationships are still captured.

```
> summary(model_reduced)$coefficients
            Estimate Std. Error  t value    Pr(>|t|)
(Intercept) 1.1086723 0.18418699   6.019276 2.788391e-09
Age          0.5561667 0.01439403  38.638710 8.142669e-178
Smokeyes    -0.6431029 0.18680655  -3.442614 6.093803e-04
Caesareanyes -0.1460278 0.13467911  -1.084265 2.786098e-01
```


#(Q20)Predict results for this regression line on the training data.

```
> ypred_reduced <- predict(model_reduced, newdata = lung)
> head(ypred_reduced)
      1      2      3      4      5      6
4.445673 10.476571  9.861312  8.895007  3.889506  7.226506
>
```

#(Q21)Calculate the mean squared error (MSE)of the training data.

```
> #(Q21)Calculate the mean squared error (MSE)of the training c
> mse <- mean((lung$LungCap - ypred_reduced)^2)
> mse
[1] 2.280169
> |
```